

DRY-RUN GUIDE | common_element.py

Source: Python\3rd_Belt\Matrix\common_element.py

CORE BEHAVIOR

Finds distinct common values of two sorted arrays with two pointers; prints -1 for an empty intersection.

REPRESENTATIVE INPUT

Arrays '[1,2,2,4,6]' and '[2,2,3,6]'.

STEP TRACE

Advance smaller pointer to 2; record 2 once; later match 6 and record it.

EXPECTED OUTPUT / RETURN

'2 6'

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt\Matrix\common_element.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Find the common elements between two sorted arrays.
```

```
002 | # Add your solution here.
```

```
003 |
```

```
004 |
```

```
005 | def find_intersection(arr1, arr2):
```

```
006 |     i, j = 0, 0
```

```
007 |     result = []
```

```
008 |
```

```
009 |     while i < len(arr1) and j < len(arr2):
```

```
010 |         if arr1[i] == arr2[j]:
```

```
011 |             if not result or result[-1] != arr1[i]:
```

```
012 |                 result.append(arr1[i])
```

```
013 |                 i += 1
```

```
014 |                 j += 1
```

```
015 |             elif arr1[i] < arr2[j]:
```

```
016 |                 i += 1
```

```
017 |             else:
```

```
018 |                 j += 1
```

```
019 |
```

```
020 |     return result
```

```
021 |
```

```
022 |
```

```
023 | if __name__ == "__main__":
```

```
024 |     arr1 = list(map(int, input().split()))
```

```
025 |     arr2 = list(map(int, input().split()))
```

```
026 |
```

```
027 |     result = find_intersection(arr1, arr2)
```

```
028 |     if not result:
```

```
029 |         print(-1)
```

```
030 |     else:
```

```
031 |         print(*result)
```

```
032 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.